Causal execution Jacobians: conditional implementation audit

EAGM 2.0 research notes; no novelty claim

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1 Timing and observable cache

At the beginning of a controller call write $c_t = (r_t, f_t, d_t)$ and $b_t = f_{t-1}$. The implemented update is

$$\bar{a}_t = \mathrm{clip}(a_t, -a_{\max}, a_{\max}), \qquad r_{t+1} = \mathrm{clip}(r_t + \bar{a}_t, \ell, u), \tag{1}$$

$$f_{t+1} = \alpha r_{t+1} + (1-\alpha) f_t, \qquad b_{t+1} = f_t, \tag{2}$$

$$v_t = \chi_{d_t=0} f_{t+1} + \chi_{d_t=1} f_t + \chi_{d_t=2} b_t. \tag{3}$$

Here χ denotes an indicator. The two-slot FIFO before the call is $[b_t, f_t]$; selection precedes advancement. Reset fills both slots with the reset target. Delay is a discrete external input, not a differentiable variable. This identity remains valid under changes between declared delays0/1/2 because the queue advances independently of delay. The imagined implementation keeps delay fixed; it does not implement an anticipated changing-delay schedule.

For fixed known $0 < \alpha < 1$ and valid preceding updates, $b_t = (f_t - \alpha r_t)/(1-\alpha)$. This is singular at $\alpha = 1$ and ill-conditioned near it. The implemented public history cache avoids that division, but requires a valid reset or actual preceding observed filter. Arbitrary initial pairs do not imply a valid FIFO.

2 Partial and propagated derivatives

On an open clipping regime let D_a and D_r be diagonal matrices of zeros/ones indicating unsaturated action and target coordinates. Set $M = D_r D_a$.

Proposition 1 *At fixed pre-call state, $\partial v_t / \partial a_t = \chi_{d_t=0} \alpha M$. Nevertheless a previous action can influence later positive-delay commands.*

Proof. Differentiate the update at fixed old states. Positive-delay selection reads only those old states. For propagation, differentiating with respect to any fixed policy parameter vector gives

$$R_{t+1} = D_r(R_t + D_a A_t), \tag{4}$$

$$F_{t+1} = \alpha R_{t+1} + (1-\alpha) F_t, \qquad B_{t+1} = F_t, \tag{5}$$

$$V_t = \chi_{d_t=0} F_{t+1} + \chi_{d_t=1} F_t + \chi_{d_t=2} B_t, \tag{6}$$

where capitals denote total parameter sensitivities. Thus V_t need not vanish at positive delay. No derivative is asserted at a clipping boundary. □

3 Controller B is a different executed-command map

Gain multiplies v_t by g. Extra lag appends $g v_t$ to an independent hidden queue, popping its oldest entry only once its length exceeds the configured lag L; during queue fill it uses the previous executed command. Denote this selection by w_t. Then

$$\lambda_t = u_{t-1} + \mathrm{clip}(w_t - u_{t-1}, -\rho, \rho), \tag{7}$$

$$s_t = \beta \lambda_t + (1-\beta) s_{t-1} \qquad (\beta > 0), \tag{8}$$

$$u_t = \mathrm{clip}(s_t, \ell, u) \qquad (\beta > 0), \tag{9}$$

while for $\beta \leq 0$ the code bypasses smoothing and clips λ_t. In particular the bypass at zero is not the mathematical limit of the positive-β recurrence. On a fixed rate-limit/output-clipping regime, with diagonal masks D_ρ, D_u,

$$\Lambda_t = D_\rho W_t + (I - D_\rho) U_{t-1}, \tag{10}$$

$$S_t = \beta \Lambda_t + (1-\beta) S_{t-1}, \quad U_t = D_u S_t \qquad (\beta > 0), \tag{11}$$

$$U_t = D_u \Lambda_t \qquad (\beta \leq 0). \tag{12}$$

The queued sensitivity W_t is gV_{t-L} after fill for constant gain/lag, with the specified startup exception. These formulas hold for policy-parameter derivatives with execution parameters fixed. They do not differentiate integer lag or the branching condition in beta. They show exactly which hidden sensitivities a memoryless nominal prior omits. A learned residual can approximate them; its success is an empirical question.

4 Code correspondence and scope

These equations follow `controllers/partial_unknown.py:step,_apply_unknown` and `eagm2/prior.py:observable_step`. The recurrent model's posterior uses observations during burn-in; imagination advances its learned transition without future physical observations. Its latent vector is not proven equivalent to the smoothing state, previous command and extra queue.

Contact dynamics are outside this algebraic controller identity. A finite-horizon gradient through physical dynamics requires differentiability of the actual discrete simulator rollout along the parameter perturbations. Remaining within one fixed active set is a sufficient local modeling restriction only when the solver map there is smooth; contact-event changes, solver branches, and clipping surfaces are not covered by this assertion. Being piecewise smooth does not justify applying a single smooth derivative across a switch. An autograd-selected subgradient at a tie is not a certified physical derivative.